

Chapter 4: Mathematical Expectation

STAT2920-1: Introduction to Probability

Mohamed Belalia, PhD

University of Windsor
Department of Mathematics and Statistics



University
of Windsor

- 1 Introduction
- 2 The Expected Value of a Random Variable
- 3 Moments about the origin
- 4 Chebyshev's Theorem
- 5 Moment generating function

- The terminology of **expectation** or **expected value** has its origin in games of chance.
- This can be illustrated as follows:

Example 1.

- Four small similar chips, numbered 1, 1, 1, and 2, respectively, are placed in a bowl and are mixed.
- A player is blindfolded and is to draw a chip from the bowl.
- If she draws one of the three chips numbered 1, she will receive one dollar.
- If she draws the chip numbered 2, she will receive two dollars.
- It seems reasonable to assume that the player has a " $\frac{3}{4}$ " claim on the \$ 1 and a " $\frac{1}{4}$ " on the \$ 2.
- Her "total claim" is $(1)(\frac{3}{4}) + 2(\frac{1}{4}) = \frac{5}{4}$
- that is, \$ 1.25.

- Thus the **expectation** of X is precisely the player's claim in this game.

Example 2.

The **roulette**^a game consists of a small ball and a wheel with **38** numbered pockets around the edge. As the wheel is spun, the ball bounces around randomly until it settles down in one of the pockets. Suppose random variable X represents the (monetary) outcome of a **\$1** bet on a single number ("straight up" bet). If the bet wins (which happens with probability $\frac{1}{38}$ in American roulette), the payoff is **\$35**; otherwise the player loses the bet. The expected profit from such a bet will be

$$\mathbb{E}[\text{gain from \$1 bet}] = -\$1 \cdot \frac{37}{38} + \$35 \cdot \frac{1}{38} = -\$ \frac{1}{19}.$$

That is, the bet of **\$1** stands to lose $-\$ \frac{1}{19}$, so its expected value is $-\$ \frac{1}{19}$.

^aThis example was taken for Wikipedia

Definition 1.

Let X be a random variable.

- If X is a discrete with and $f(x)$ is the value of its probability distribution at x , the expected value of X is

$$\mathbb{E}(X) = \sum_x x \cdot f(x).$$

- Correspondingly If X is a continuous and $f(x)$ is the value of its probability density at x , the expected value of X is

$$\mathbb{E}(X) = \int_{-\infty}^{+\infty} x \cdot f(x) dx.$$

Example 3.

Let X have a pdf

$$f(x) = \begin{cases} 4x^3 & \text{if } 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $\mathbb{E}(X)$.

Example 3.

Let X have a pdf

$$f(x) = \begin{cases} 4x^3 & \text{if } 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $\mathbb{E}(X)$.

Solution:

$$\mathbb{E}(X) = \int_0^1 x \cdot f(x) \, dx = \int_0^1 4x^4 \, dx = \left. \frac{4x^5}{5} \right|_0^1 = \frac{4}{5}.$$

Example 4.

Let X be a r.v. whose *pmf* is

x	0	1	Total
$p(x)$	$1 - p$	p	1

where $0 < p < 1$. Find $\mathbb{E}(X)$

Example 4.

Let X be a r.v. whose *pmf* is

x	0	1	Total
$p(x)$	$1 - p$	p	1

where $0 < p < 1$. Find $\mathbb{E}(X)$

Solution:

$$\mathbb{E}(X) = \sum_{x=0}^1 xp(x) = 0 \times p(0) + 1 \times p(1) = p.$$

Example 5.

Let X be a r.v. whose *pmf* is

x	1	2	3	4
$p(x)$	$\frac{4}{10}$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{2}{10}$

Find $\mathbb{E}(X)$.

Example 5.

Let X be a r.v. whose *pmf* is

x	1	2	3	4
$p(x)$	$\frac{4}{10}$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{2}{10}$

Find $\mathbb{E}(X)$.

Solution:

$$\begin{aligned}
 \mathbb{E}(X) &= \sum_{x=1}^4 xp(x) = 1.p(1) + 2.p(2) + 3.p(3) + 4.p(4) \\
 &= 1.\frac{4}{10} + 2.\frac{1}{10} + 3.\frac{3}{10} + 4.\frac{2}{10} \\
 &= \frac{23}{10}.
 \end{aligned}$$

Example 6.

A lot of 12 television sets includes 2 with white cords. If 3 of the sets are chosen at random for shipment to a hotel, how many sets with white cords can the shipper expect to send to the hotel?

Example 6.

A lot of 12 television sets includes 2 with white cords. If 3 of the sets are chosen at random for shipment to a hotel, how many sets with white cords can the shipper expect to send to the hotel?

Solution :

- Since x of the 2 sets with white cords and $3 - x$ of the 10 other sets can be chosen in $\binom{2}{x} \binom{10}{3-x}$ ways, 3 of the 12 sets can be chosen in $\binom{12}{3}$ ways, and these $\binom{12}{3}$ possibilities are presumably equiprobable,
- The *pmf* of X , the number of sets with white cords shipped to the hotel, is given by

$$f(x) = \frac{\binom{2}{x} \binom{10}{3-x}}{\binom{12}{3}} \quad x = 0, 1, 2$$

or, in tabular form,

x	0	1	2
$f(x)$	$\frac{6}{11}$	$\frac{9}{22}$	$\frac{1}{22}$

- Now

$$\mathbb{E}[X] = 0 \cdot \frac{6}{11} + 1 \cdot \frac{9}{22} + 2 \cdot \frac{1}{22} = \frac{1}{2}$$

Example 7.

Certain coded measurements of the pitch diameter of threads of a fitting have the probability density

$$f(x) = \begin{cases} \frac{4}{\pi(1+x^2)} & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the expected value of this random variable.

Example 7.

Certain coded measurements of the pitch diameter of threads of a fitting have the probability density

$$f(x) = \begin{cases} \frac{4}{\pi(1+x^2)} & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the expected value of this random variable.

Solution :

- Using Definition 1, we get

$$\begin{aligned} \int_0^1 x \cdot \frac{4}{\pi(1+x^2)} dx &= \frac{2}{\pi} \int_0^1 \frac{2x}{\pi(1+x^2)} dx \\ &= \frac{2 \ln 2}{\pi} = 0.4413. \end{aligned}$$

Theorem 1.

- If X is a discrete random variable and $f(x)$ is the value of its probability distribution at x , the expected value of $g(X)$ is given by

$$\mathbb{E}[g(X)] = \sum_x g(x) \cdot f(x)$$

- Correspondingly, if X is a continuous random variable and $f(x)$ is the value of its probability density at x , the expected value of $g(X)$ is given by

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) \cdot f(x) \, dx$$

Since a more general proof is beyond the scope of this course, we shall prove this theorem here only for the case where X is discrete and has a finite range. Since $y = g(x)$ does not necessarily define a one-to-one correspondence, suppose that $g(x)$ takes on the value g_i when x takes on the values $x_{i1}, x_{i2}, \dots, x_{in_i}$. Then, the probability that $g(X)$ will take on the value g_i is

$$\mathbb{P}[g(X) = g_i] = \sum_{j=1}^{n_i} f(x_{ij})$$

and if $g(x)$ takes on the values $g_1, g_2, \dots, g_m$, it follows that

$$\begin{aligned} \mathbb{E}[g(X)] &= \sum_{i=1}^m g_i \cdot \mathbb{P}[g(X) = g_i] = \sum_{i=1}^m g_i \cdot \sum_{j=1}^{n_i} f(x_{ij}) \\ &= \sum_{i=1}^m \sum_{j=1}^{n_i} g_i \cdot f(x_{ij}) \\ &= \sum_x g(x) \cdot f(x). \end{aligned}$$

where the summation extends over all values of X .

Example 8.

Let X be a discrete r.v. with *pmf* is

$$p(x) = \begin{cases} \frac{x}{6} & \text{for } x = 1, 2, 3 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $\mathbb{E}(X^3)$.

Example 8.

Let X be a discrete r.v. with pmf is

$$p(x) = \begin{cases} \frac{x}{6} & \text{for } x = 1, 2, 3 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $\mathbb{E}(X^3)$.

Solution:

$$\begin{aligned} \mathbb{E}(X^3) &= \sum_{x=1}^3 x^3 p(x) = 1^3 \cdot p(1) + 2^3 \cdot p(2) + 3^3 \cdot p(3) \\ &= 1 \cdot \frac{1}{6} + 8 \cdot \frac{2}{6} + 27 \cdot \frac{3}{6} \\ &= \frac{1 + 16 + 81}{6} = \frac{98}{6} = \frac{49}{3}. \end{aligned}$$

Example 9.

Let X have the pdf

$$f(x) = \begin{cases} 2(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $\mathbb{E}(3X^2)$.

Example 9.

Let X have the pdf

$$f(x) = \begin{cases} 2(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $\mathbb{E}(3X^2)$.

Solution:

$$\mathbb{E}(3X^2) = \int_0^1 3x^2 f(x) \, dx = \int_0^1 6x^2(1-x) \, dx = \left(\frac{6x^3}{3} - \frac{6x^4}{4} \right) \bigg|_0^1 = \frac{1}{2}.$$

Example 10.

If X is the number of points rolled with a balanced die, find the expected value of

$$g(X) = 2X^2 + 1.$$

Example 10.

If X is the number of points rolled with a balanced die, find the expected value of

$$g(X) = 2X^2 + 1.$$

Solution: Since each possible outcome has the probability $\frac{1}{6}$, we get

$$\begin{aligned}\mathbb{E}[g(X)] &= \sum_{x=1}^6 (2x^2 + 1) \cdot \frac{1}{6} \\ &= (2 \cdot 1^2 + 1) \cdot \frac{1}{6} + \dots + (2 \cdot 6^2 + 1) \cdot \frac{1}{6} \\ &= \frac{94}{3}.\end{aligned}$$

Exercise 1.

If X has the probability density

$$f(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the expected value of $g(X) = e^{3X/4}$.

Exercise 2.

If X has the probability density

$$f(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the expected value of $g(X) = e^{3X/4}$.

Solution 2.

According to Theorem 1, we have

$$\begin{aligned} \mathbb{E} \left[e^{3X/4} \right] &= \int_0^{\infty} e^{3x/4} \cdot e^{-x} \, dx \\ &= \int_0^{\infty} e^{-x/4} \, dx \\ &= \left(-4e^{-x/4} \right) \Big|_0^{\infty} \\ &= 4. \end{aligned}$$

Theorem 2.

- If a and b are constants, then

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$$

Theorem 2.

- If a and b are constants, then

$$\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$$

Proof.

$$\begin{aligned}\mathbb{E}[aX + b] &= \int_{-\infty}^{\infty} (ax + b)f(x)dx = a \underbrace{\int_{-\infty}^{\infty} xf(x)dx}_{= \mathbb{E}[X]} + b \underbrace{\int_{-\infty}^{\infty} f(x)dx}_{= 1} \\ &= a\mathbb{E}[X] + b.\end{aligned}$$



- If we set $a = 0$ or $b = 0$, we can state the following corollaries to Theorem 2.

Corollary 1.

- If a is a constants, then

$$\mathbb{E}(aX) = a\mathbb{E}(X).$$

- If we set $a = 0$ or $b = 0$, we can state the following corollaries to Theorem 2.

Corollary 1.

- If a is a constants, then

$$\mathbb{E}(aX) = a\mathbb{E}(X).$$

Corollary 2.

- If b is a constants, then

$$\mathbb{E}(b) = b.$$

Theorem 3.

- If $c_1, c_2, \dots$ and c_n are constants, then

$$\mathbb{E} \left[\sum_{i=1}^n c_i g_i(X) \right] = \sum_{i=1}^n c_i \mathbb{E} [g_i(X)]$$

Proof.

According to Theorem 1 with $g(X) = \sum_{i=1}^n c_i g_i(X)$, we get

$$\begin{aligned}\mathbb{E} \left[\sum_{i=1}^n c_i g_i(X) \right] &= \sum_x \left[\sum_{i=1}^n c_i g_i(x) \right] f(x) \\ &= \sum_{i=1}^n \sum_x c_i g_i(x) f(x) \\ &= \sum_{i=1}^n c_i \sum_x g_i(x) f(x) \\ &= \sum_{i=1}^n c_i \mathbb{E} [g_i(X)]\end{aligned}$$



Exercise 3.

If the probability density of X is given by

$$f(x) = \begin{cases} 2(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

1. Show that

$$\mathbb{E}(X^r) = \frac{2}{(r+1)(r+2)}, \quad \text{for } r > 0$$

2. Use this result to evaluate

$$\mathbb{E}[(2X+1)^2]$$

Solution 3.

1.

$$\begin{aligned}\mathbb{E}(X^r) &= \int_0^1 x^r f(x) dx = \int_0^1 x^r 2(1-x) dx = 2 \int_0^1 [x^r - x^{r+1}] dx \\ &= 2 \left[\frac{x^{r+1}}{r+1} - \frac{x^{r+2}}{r+2} \right] \bigg|_0^1 \\ &= \frac{2}{(r+1)(r+2)}\end{aligned}$$

2. Using (1), we get

$$\begin{aligned}\mathbb{E}[(2X+1)^2] &= \mathbb{E}[4X^2 + 4X + 1] = 4\mathbb{E}[X^2] + 4\mathbb{E}[X] + 1 \\ &= 4 \cdot \frac{2}{3 \cdot 4} + 4 \cdot \frac{2}{2 \cdot 3} + 1 \\ &= 3.\end{aligned}$$



Exercise 4.

Show that

$$\mathbb{E}[(aX + b)^n] = \sum_{i=1}^n \binom{n}{i} a^{n-i} b^i \mathbb{E}(X^{n-i})$$

Solution 4.

Since $(ax + b)^n = \sum_{i=1}^n \binom{n}{i} (ax)^{n-i} b^i$, it follows that

$$\begin{aligned} \mathbb{E}[(aX + b)^n] &= \mathbb{E}\left[\sum_{i=1}^n \binom{n}{i} (aX)^{n-i} b^i\right] \\ &= \sum_{i=1}^n \binom{n}{i} a^{n-i} b^i \mathbb{E}[X^{n-i}] \end{aligned}$$



Exercise 5.

Suppose that a random variable X is equal to the number of hits obtained by a certain baseball player in his next 3 bats. If

$$P(X = 1) = 0.3$$

$$P(X = 2) = 0.2$$

$$P(X = 0) = 3P(X = 3)$$

find $\mathbb{E}[X]$.

Solution 5.

$$\sum_{i=0}^3 P(X = i) = 1$$

$$\therefore 0.3 + 0.2 + 4P(X = 3) = 1$$

$$\therefore 4P(X = 3) = 0.5$$

$$\therefore P(X = 3) = 0.125$$

$$\therefore P(X = 0) = 0.375$$

Therefore,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i=0}^3 iP(X = i) \\ &= (0)(0.375) + (1)(0.3) + (2)(0.2) + (3)(0.125) \\ &= 0 + 0.3 + 0.4 + 0.375 \\ &= 1.075\end{aligned}$$

Exercise 6.

Suppose that X takes on one of the values 0, 1, or 2. If for some constant c ,

$$P(X = i) = cP(X = i - 1)$$

for $i = 1$ and $i = 2$, find $\mathbb{E}[X]$ in terms of c .

Solution 6.

$$\sum_{i=0}^2 P(X = i) = 1$$

$$\therefore P(X = 0) + cP(X = 0) + c^2P(X = 0) = 1$$

$$\therefore P(X = 0) = \frac{1}{1 + c + c^2}$$

Therefore,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i=0}^2 iP(X = i) \\ &= (0) \left(\frac{1}{1 + c + c^2} \right) + (1) \left(\frac{c}{1 + c + c^2} \right) + (2) \left(\frac{c^2}{1 + c + c^2} \right) \\ &= \frac{c + 2c^2}{1 + c + c^2}\end{aligned}$$

Definition 2.

The r^{th} moment about the origin of a random variable X , denoted by μ'_r , is the expected value of X^r ; symbolically

$$\mu'_r = \mathbb{E}(X^r) = \sum_x x^r \cdot f(x)$$

for $r = 0, 1, 2, 3, \dots$, when X is discrete, and

$$\mu'_r = \mathbb{E}(X^r) = \int_{-\infty}^{\infty} x^r \cdot f(x) \, dx$$

when X is continuous.

Definition 3.

μ'_1 , is called the **mean** of the distribution of X , or simply the mean of X , and it is denoted simply by μ .

Definition 4.

The r^{th} moment about the mean of a random variable X , denoted by μ_r , is the expected value of $(X - \mu)^r$; symbolically

$$\mu_r = \mathbb{E} [(X - \mu)^r] = \sum_x (x - \mu)^r \cdot f(x)$$

for $r = 0, 1, 2, 3, \dots$, when X is discrete, and

$$\mu_r = \mathbb{E} [(X - \mu)^r] = \int_{-\infty}^{\infty} (x - \mu)^r \cdot f(x) \, dx$$

when X is continuous.

Definition 5.

μ_2 , is called the **variance** of the distribution of X , or simply the variance of X , and it is denoted simply by σ^2 , σ_X^2 , $\text{Var}(X)$ or $V(X)$. The positive square root of the variance, σ , is called the **standard deviation** of X .

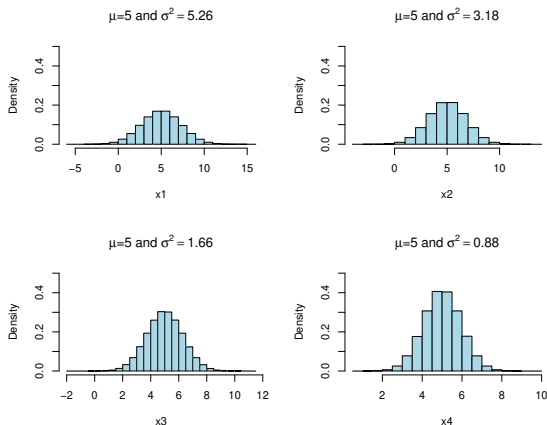


Figure 1: Distributions with different dispersions

Theorem 4.

$$\sigma^2 = \mu'_2 - \mu^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2$$

Theorem 4.

$$\sigma^2 = \mu'_2 - \mu^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2$$

Proof.

$$\begin{aligned}\sigma^2 &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}(X^2 - 2\mu X + \mu^2) \\ &= \mathbb{E}(X^2) - 2\mu \mathbb{E}(X) \mathbb{E}(\mu^2) \\ &= \mathbb{E}(X^2) - 2\mu \cdot \mu + \mu^2 \\ &= \mu'_2 - \mu^2\end{aligned}$$



Example 11.

Let X have the following pdf

$$f(x) = \begin{cases} 2x & \text{if } x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}.$$

Find $\mathbb{E}(X)$ and $\mathbb{Var}(X)$.

Example 11.

Let X have the following pdf

$$f(x) = \begin{cases} 2x & \text{if } x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}.$$

Find $\mathbb{E}(X)$ and $\mathbb{V}\text{ar}(X)$.

Solution: One can get easily $\mathbb{E}[X] = 2/3$. Then, we have

$$\mathbb{E}[X^2] = \int_0^1 x^2(2x) \, dx = \left[\frac{2x^4}{4} \right]_{x=0}^1 = \frac{1}{2}(1^4 - 0^4) = 1/2,$$

hence,

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1}{2} - \frac{4}{9} = \frac{9-8}{18} = \frac{1}{18}.$$

Example 12.

Use Theorem 4 to calculate the variance of X , representing the number of points rolled with a balanced die.

Example 12.

Use Theorem 4 to calculate the variance of X , representing the number of points rolled with a balanced die.

Solution:

- First we compute

$$\mu = \mathbb{E}(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}$$

- Now,

$$\mu'_2 = \mathbb{E}(X^2) = 1^2 \cdot \frac{1}{6} + 2^2 \cdot \frac{1}{6} + 3^2 \cdot \frac{1}{6} + 4^2 \cdot \frac{1}{6} + 5^2 \cdot \frac{1}{6} + 6^2 \cdot \frac{1}{6} = \frac{91}{6}$$

- It follows that

$$\sigma^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}.$$

Example 13.

With reference to Example 7, find the standard deviation of the random variable X .

Example 13.

With reference to Example 7, find the standard deviation of the random variable X .

Solution: In Example 7 we showed that $\mu = \mathbb{E}(X) = 0.4413$. Now

$$\begin{aligned}\mu'_2 = \mathbb{E}(X^2) &= \frac{4}{\pi} \int_0^1 \frac{x^2}{1+x^2} dx \\ &= \frac{4}{\pi} \int_0^1 \left(1 - \frac{1}{1+x^2}\right) dx \\ &= \frac{4}{\pi} [x - \arctan(x)]_0^1 = \frac{4}{\pi} \left[1 - \frac{\pi}{4}\right] \\ &= \frac{4}{\pi} - 1 = 0.2732.\end{aligned}$$

and it follows that

$$\sigma^2 = 0.2732 - (0.4413)^2 = 0.0785.$$

and $\sigma = \sqrt{0.0785} = 0.2802$.

- The following is another theorem that is of importance in work connected with standard deviations or variances.

Theorem 5.

If X has variance σ^2 , then

$$\mathbb{V}ar(aX + b) = a^2 \mathbb{V}ar(X) = a^2 \sigma^2.$$

- The following is another theorem that is of importance in work connected with standard deviations or variances.

Theorem 5.

If X has variance σ^2 , then

$$\text{Var}(aX + b) = a^2 \text{Var}(X) = a^2 \sigma^2.$$

Proof.

$$\begin{aligned}\text{Var}(aX + b) &= \mathbb{E} \left[aX + b - (a\mathbb{E}[X] + b) \right]^2 \\ &= \int_{-\infty}^{\infty} \left[ax + b - (a\mathbb{E}[X] + b) \right]^2 f(x) \, dx \\ &= \int_{-\infty}^{\infty} a^2 (x - \mathbb{E}[X])^2 f(x) \, dx \\ &= a^2 \text{Var}(X).\end{aligned}$$



Theorem 6.

If μ and σ are the mean and the standard deviation of a random variable X , then for any positive constant k the probability is at least $1 - \frac{1}{k^2}$ that X will take on a value within k standard deviations of the mean; symbolically,

$$\mathbb{P}(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}, \quad \sigma \neq 0$$

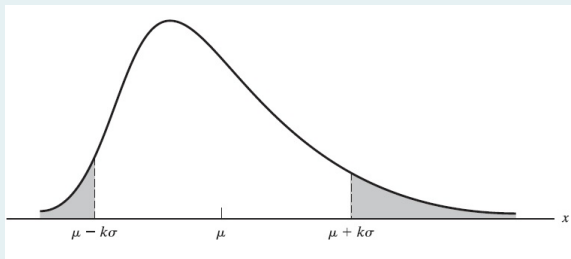


Figure 2: Diagram for Chebyshev's theorem.

According to Definitions 3 and 5, we write

$$\sigma^2 = \mathbb{E} [(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) \, dx$$

Then, dividing the integral into three parts as shown in Figure 2, we get

$$\sigma^2 = \int_{-\infty}^{\mu - k\sigma} (x - \mu)^2 \cdot f(x) \, dx + \int_{\mu - k\sigma}^{\mu + k\sigma} (x - \mu)^2 \cdot f(x) \, dx + \int_{\mu + k\sigma}^{\infty} (x - \mu)^2 \cdot f(x) \, dx$$

Since the integrand $(x - \mu)^2 \cdot f(x)$ is nonnegative, we can form the inequality

$$\sigma^2 \geq \int_{-\infty}^{\mu - k\sigma} (x - \mu)^2 \cdot f(x) \, dx + \int_{\mu + k\sigma}^{\infty} (x - \mu)^2 \cdot f(x) \, dx$$

by deleting the second integral. Therefore, since $(x - \mu)^2 \geq k^2\sigma^2$ for $\mu - k\sigma$ or $x \geq \mu + k\sigma$ it follows that

$$\sigma^2 \geq \int_{-\infty}^{\mu - k\sigma} k^2\sigma^2 \cdot f(x) \, dx + \int_{\mu + k\sigma}^{\infty} k^2\sigma^2 \cdot f(x) \, dx$$

and hence

$$\frac{1}{k^2} \geq \int_{-\infty}^{\mu - k\sigma} f(x) \, dx + \int_{\mu + k\sigma}^{\infty} f(x) \, dx$$

provided $\sigma^2 \neq 0$. Since the sum of the two integrals on the right-hand side is the probability that X will take on a value less than or equal to $\mu - k\sigma$ or greater than or equal to $\mu + k\sigma$, we have thus shown that

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}, \quad \sigma \neq 0$$

and it follows that

$$\mathbb{P}(|X - \mu| \geq k\sigma) \geq 1 - \frac{1}{k^2}, \quad \sigma \neq 0$$

Example 14.

If the probability density of X is given by

$$f(x) = \begin{cases} 630x^4(1-x)^4 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the probability that it will take on a value within two standard deviations of the mean and compare this probability with the lower bound provided by Chebyshev's theorem.

Example 14.

If the probability density of X is given by

$$f(x) = \begin{cases} 630x^4(1-x)^4 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the probability that it will take on a value within two standard deviations of the mean and compare this probability with the lower bound provided by Chebyshev's theorem.

Solution: Straightforward integration shows that $\mu = \frac{1}{2} = 0.5$ and $\sigma^2 = \frac{1}{44}$, so that $\sigma = \sqrt{1/44}$ or approximately 0.15. Thus, the probability that X will take on a value within two standard deviations of the mean is the probability that it will take on a value between 0.20 and 0.80, that is,

$$\mathbb{P}(0.20 < X < 0.80) = \int_{0.20}^{0.80} 630x^4(1-x)^4 dx = 0.96$$

Observe that the statement “the probability is 0.96” is a much stronger statement than “the probability is at least 0.75,” which is provided by Chebyshev's theorem.

Definition 6.

The moment generating function of a random variable X , where it exists, is given by

$$M_X(t) = \mathbb{E} \left[e^{tX} \right] = \sum_x e^{tx} \cdot f(x)$$

when X is discrete, and

$$M_X(t) = \mathbb{E} \left[e^{tX} \right] = \int_{-\infty}^{\infty} e^{tx} \cdot f(x) \, dx$$

when X is continuous.

- To explain why we refer to this function as a “moment-generating” function, let us substitute for e^{tx} its Maclaurin’s series expansion, that is,

$$e^{tx} = 1 + tx + \frac{t^2 x^2}{2!} + \frac{t^3 x^3}{3!} + \dots + \frac{t^r x^r}{r!} + \dots$$

- For the discrete case, we thus get

$$\begin{aligned} M_X(t) &= \sum_x \left[1 + tx + \frac{t^2 x^2}{2!} + \frac{t^3 x^3}{3!} + \dots + \frac{t^r x^r}{r!} + \dots \right] f(x) \\ &= \sum_x f(x) + t \cdot \sum_x x f(x) + \frac{t^2}{2!} \cdot \sum_x x^2 f(x) + \dots + \frac{t^r}{r!} \cdot \sum_x x^r f(x) + \dots \\ &= 1 + \mu \cdot t + \mu'_2 \cdot \frac{t^2}{2!} + \dots + \mu'_r \cdot \frac{t^r}{r!} + \dots \end{aligned}$$

- and it can be seen that in the Maclaurin’s series of the moment-generating function of X the coefficient of $\frac{t^r}{r!}$ is μ'_r the r th moment about the origin.
- In the continuous case, the argument is the same.

Example 15.

- Let X be a random variable whose pdf is

$$f_X(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find the mgf of X .

Example 15.

- Let X be a random variable whose pdf is

$$f_X(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find the mgf of X .

Solution: By definition

$$\begin{aligned} M_X(t) &= \mathbb{E} \left[e^{tX} \right] = \int_0^{\infty} e^{tx} e^{-x} dx \\ &= \int_0^{\infty} e^{-x(1-t)} dx \\ &= \begin{cases} \frac{1}{1-t} & \text{if } t < 1 \\ +\infty & \text{if } t \geq 1. \end{cases} \end{aligned}$$

Theorem 7.

$$\left. \frac{d^r M_X(t)}{dt^r} \right|_{t=0} = \mu'_r$$

Theorem 7.

$$\left. \frac{d^r M_X(t)}{dt^r} \right|_{t=0} = \mu'_r$$

Example 16.

Given that X has the probability distribution

$$f(x) = \begin{cases} \frac{1}{8} \binom{3}{x} & \text{for } x = 0, 1, 2, 3 \\ 0 & \text{elsewhere} \end{cases}$$

find the moment-generating function of this random variable and use it to determine μ'_1 and μ'_2 .

Solution: In accordance with Definition 6,

$$\begin{aligned}M_X(t) &= \mathbb{E} \left[e^{tX} \right] = \frac{1}{8} \cdot \sum_{x=0}^3 e^{tx} \binom{3}{x} \\&= \frac{1}{8} (1 + 3e^t + 3e^{2t} + e^{3t}) \\&= \frac{1}{8} (1 + e^t)^3.\end{aligned}$$

Then by Theorem 7

$$\mu'_1 = M'_X(0) = \left. \frac{3}{8} e^t (1 + e^t)^2 \right|_{t=0} = \frac{3}{2}.$$

and

$$\mu'_2 = M''_X(0) = \left. \frac{3}{4} e^{2t} (1 + e^t) + \frac{3}{8} e^t (1 + e^t)^2 \right|_{t=0} = 3.$$

Theorem 8.

If a and b are constants, then

1.

$$M_{X+a}(t) = e^{at} \cdot M_X(t);$$

2.

$$M_{bX}(t) = M_X(bt);$$

3.

$$M_{\frac{X+a}{b}}(t) = \mathbb{E} \left[e^{\left(\frac{X+a}{b}\right)t} \right] = e^{\frac{a}{b}t} \cdot M_X \left(\frac{t}{b} \right).$$

1.

$$M_{X+a}(t) = \mathbb{E} \left[e^{t(X+a)} \right] = \mathbb{E} \left[e^{tX} \cdot e^{ta} \right] = e^{ta} \mathbb{E} \left[e^{tX} \right] = e^{ta} M_X(t).$$

2.

$$M_{bX}(t) = \mathbb{E} \left[e^{tbX} \right] = M_X(bt).$$

3.

$$\begin{aligned} M_{\frac{X+a}{b}}(t) &= \mathbb{E} \left[e^{\left(\frac{X+a}{b} \right) t} \right] = \mathbb{E} \left[e^{\frac{ta}{b}} \cdot e^{\frac{tX}{b}} \right] \\ &= e^{\frac{ta}{b}} \cdot \mathbb{E} \left[e^{\frac{tX}{b}} \right] = e^{\frac{a}{b}t} \cdot M_X \left(\frac{t}{b} \right). \end{aligned}$$

Exercise 7.

If X has the following pdf

$$f(x) = \frac{1}{2}e^{-|x|}, \quad \text{for } -\infty < x < \infty$$

find $M_X(t)$

Solution 7.

$$\begin{aligned}
M_X(t) &= \mathbb{E} \left[e^{tX} \right] = \int_{-\infty}^{\infty} e^{tx} f(x) \, dx \\
&= \underbrace{\frac{1}{2} \int_{-\infty}^0 e^{tx} e^x \, dx}_{\text{put } y=-x} + \frac{1}{2} \int_0^{\infty} e^{tx} e^{-x} \, dx \\
&= \frac{1}{2} \int_0^{\infty} e^{-ty} e^{-y} \, dy + \frac{1}{2} \int_0^{\infty} e^{tx} e^{-x} \, dx \\
&= \frac{1}{2} \int_0^{\infty} e^{-(1+t)y} \, dy + \frac{1}{2} \int_0^{\infty} e^{-(1-t)x} \, dx \\
&= -\frac{1/2}{(1+t)} e^{-(1+t)y} \Big|_0^{\infty} + -\frac{1/2}{(1-t)} e^{-(1-t)x} \Big|_0^{\infty} \\
&= \frac{1}{2} \left[\frac{1-t+1+t}{(1+t)(1-t)} \right] = \frac{1}{1-t^2}, \quad \text{for } |t| < 1.
\end{aligned}$$

Exercise 8.

If X has the following mgf

$$M_X(t) = e^{3t+8t^2}$$

find the mgf of $Z = \frac{1}{4}(X - 3)$

Solution 8.

According to (3) of Theorem 8, with $b = 4$ and $a = -3$, we get

$$\begin{aligned} M_Z(t) &= M_{\frac{X-3}{4}}(t) = e^{-\frac{3}{4}t} \cdot M_X\left(\frac{t}{4}\right) \\ &= e^{-\frac{3}{4}t} \cdot e^{\frac{3t}{4} + \frac{8t^2}{16}} \\ &= e^{\frac{t^2}{2}} \end{aligned}$$

Exercise 9.

Find the moment-generating function of the discrete random variable X that has the probability distribution

$$f(x) = 2 \left(\frac{1}{3} \right)^x, \quad \text{for } x = 1, 2, 3, \dots$$

and use it to determine the values of μ'_1 and μ'_2 .

Solution 9.

- The mgf can be derived as

$$M_X(t) = \sum_x e^{tx} 2 \left(\frac{1}{3}\right)^x = 2 \sum_{x=1}^{\infty} \left(\frac{e^t}{3}\right)^x = 2 \frac{e^t/3}{1 - e^t/3} = \frac{2e^t}{3 - e^t}$$

- The first and second derivatives of $M_X(t)$ are

$$M'_X(t) = \frac{(3 - e^t)2e^t - 2e^t(-e^t)}{(3 - e^t)^2} = \frac{6e^t}{(3 - e^t)^2}$$

$$M''_X(t) = \frac{(3 - e^t)^2 6e^t - 6e^t \cdot 2(3 - e^t)(-e^t)}{(3 - e^t)^4}$$

- Thus,

$$\mu'_1 = M'_X(0) = \frac{3}{2} \quad \text{and} \quad \mu'_2 = M''_X(0) = \frac{24 - 12 \cdot 2(-1)}{16} = 3$$

- Note that the variance can be easily obtained, as follow

$$\sigma^2 = \mu'_2 - (\mu'_1)^2 = 3 - \frac{9}{4} = \frac{3}{4}$$

Exercise 10.

1. Let Z be a random variable whose pdf is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \quad z \in \mathbb{R}.$$

Find the mgf of Z . We denote $Z \sim \mathcal{N}(0, 1)$

2. Let X be a random variable whose pdf is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Find the mgf of X . We denote $X \sim \mathcal{N}(\mu, \sigma^2)$.

Solution 10.

1.

$$\begin{aligned}\mathbb{E}\left[e^{tZ}\right] &= \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \\ &= e^{t^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2} + \frac{2tz}{2} - \frac{t^2}{2}\right) dz \\ &= \exp\left(\frac{t^2}{2}\right) \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(z-t)^2}{2}\right) dz}_{=1} = \exp\left(\frac{t^2}{2}\right).\end{aligned}$$

2. For the second, note that $X \stackrel{\text{law}}{=} \mu + \sigma Z$, then

$$M_X(t) = E[e^{t(\mu + \sigma Z)}] = e^{t\mu} E[e^{(t\sigma)Z}] = \exp\left(t\mu + \frac{(t\sigma)^2}{2}\right).$$

Main textbook:

- John E. Freund's. *Mathematical Statistics With Applications* Ed: 8 (Required).

Suggested textbooks:

- Ross, S., 2009. *A first course in probability*. Upper Saddle River, 6.
- Hossein Pishro-Nik. *Probability, Statistics and Random Processes* [here](#).

Some useful websites:

- Society of Actuaries, [Exam of probability, sample questions](#).
- Society of Actuaries, [Exam of probability, sample solutions](#).